

## ABSTRACT

This project examines historical climate data and recent weather trends to determine whether or not weather patterns have become more abnormal over time. Using data from the National Centers for Environmental Information (NOAA), we applied both classification and regression models to assess changes in climate patterns. Classification models were trained on older climate data to predict whether recent weather patterns are significantly different from historical norms, while regression models were used to predict temperature and precipitation based on past trends. High misclassification rates on recent data seemed to suggest that climates are becoming more similar to one another based on recent trends. However, the regression models were limited in their predictive power and this will be examined further in the discussion. The effectiveness of the models was evaluated using various performance metrics, including F1 scores, Precision, Recall, and error metrics such as Mean Squared Error (MSE) for regression. We hope to provide insights into how climate patterns may have shifted over time and whether these changes are consistent across different regions.

## I. INTRODUCTION

### *A. Statement Problem*

Climate change has led to increasing variability in weather patterns worldwide, raising concerns about the stability of climate zones and the predictability of local conditions. While global warming is well-documented, less is known about how distinct climate regions may be converging or diverging in their behavior over time. Our project focuses on these questions at a national scale: Are different parts of the United States experiencing more similar weather patterns than before? Can we still reliably predict expected temperature and precipitation levels for specific regions? And more broadly, are abnormal or extreme weather events becoming more frequent?

To answer these questions, we use historical weather data provided by the National Centers for Environmental Information (NOAA), a government agency that has tracked climate trends in the U.S. for over a century. Specifically, we analyze the Local Climatological Data and Normals Annual/Seasonal datasets, which contain detailed regional information on temperature, precipitation, and long-term averages [1].

Given the volume and complexity of climate data, machine learning (ML) offers a powerful toolkit for uncovering subtle trends and identifying shifts that may not be apparent

through traditional statistical methods. ML models can detect non-linear relationships and patterns that evolve over time, making them ideal for assessing whether current weather patterns deviate meaningfully from historical norms. In our project, we apply classification models to determine whether recent weather data aligns with past trends, and regression models to predict temperature and precipitation based on historical values. We train our models on pre-2000 data and test them on post-2010 data to evaluate changes over time.

### *B. Importance*

Understanding how climate patterns are shifting is critical for decision-making across multiple sectors, including agriculture, urban planning, biodiversity protection, infrastructure development, and disaster preparedness. For example, if regions like California begin to experience rainfall patterns more typical of the Midwest, this could drastically alter irrigation strategies and crop planning. Similarly, infrastructure designed based on historical weather patterns may no longer be resilient in the face of emerging extremes. In Boston, for instance, the Seaport District—a billion dollar development that was built on artificial land along the waterfront—may face the risk of being underwater in years to come due to rising sea levels [2]. Industries such as insurance and energy also rely on climate stability to forecast risk and demand. By identifying and quantifying regional shifts in weather behavior, this project aims to inform more adaptive, data-driven approaches to policy and planning. We hope our findings can contribute to the broader effort of updating environmental models and sustainability strategies in the face of ongoing climate change.

### *C. Background Information*

Research over the past several decades has confirmed a global trend of rising temperatures, increased climate variability, and more frequent extreme weather events. According to the Intergovernmental Panel on Climate Change (IPCC), the average global surface temperature has increased by approximately 1.1°C since the pre-industrial era, with significant regional differences in how this warming is experienced [3]. In the United States, studies have shown increases in both the frequency and intensity of heatwaves, shifts in precipitation patterns, and longer wildfire seasons, particularly in the western states [3].

Several studies have also explored how climate zones are shifting geographically. For instance, a study by Loarie et al. (2009) introduced the concept of "climate velocity"—the rate at which species must move to stay within their preferred climate conditions—as a metric for understanding regional climate transformation [4]. This research suggests that climate zones in many areas are moving northward or to higher elevations, potentially leading to biome convergence over time.

Our project builds on this body of work by using NOAA's long-term climate datasets and machine learning models to investigate changes in regional weather predictability. By focusing on both classification and regression tasks, we aim to not only identify if recent weather behaves differently from historical norms but also evaluate whether this divergence is occurring unevenly

across different parts of the country. This approach allows us to contribute a new perspective to the understanding of localized climate change through the lens of predictive modeling.

## II. METHODOLOGY

To accurately represent each region of the United States, weather stations situated in different climates were picked from NOAA data. The cities and their associated climates are as follows: Miami, Florida representing the country's tropical climate; Boston, Massachusetts, representing a humid subtropical climate; Winslow, Arizona representing a dry, temperate desert environment; San Francisco representing a Mediterranean, coastal climate; and Chicago, Illinois representing a humid, continental environment. These cities were also picked due to their spread covering multiple regions of the United States to assess if climate change is reflected across the continent or only regional areas.

To investigate these patterns of climate change, two datasets were retrieved from NOAA: a dataset representing the past and another reflecting the present day. The past dataset, from years 1950 to 1960, included daily information regarding features such as dry bulb, departure from average temperature, heating and cooling degree days, wind speed and direction, precipitation, snow fall, and snow depth. A present dataset was constructed using similar features, and encompassed the years between 2022 and 2025. Missing data, which appeared to primarily represent the absence of a type of weather (e.g. snowfall), was replaced with zeroes, and trace amounts, represented by 'T', were replaced with 0.005. These substitutions better allowed for the models to handle these datasets while keeping the significance of the missing values.

After extracting this data, three different types of models were created to evaluate climate change patterns: a binary classification logistic regression model, a climate/city random forest classifier, and a linear regression model to predict various features. General pre-processing steps remained similar for all three of these models and is discussed in the Code section. However, important distinctions between the models will be highlighted below.

Due to the model's aptitude for binary classification, logistic regression was picked for the first classification model. In order to evaluate climate change, a separate boolean column titled 'Abnormal Temperature' identified if any of the features were outside the norm. Any value in a row that was outside of the feature's first and third quantile were deemed abnormal, resulting in 15,422 instances of normal classes and 8,082 instances of abnormal across both datasets. Multiple models with different regularization values were fitted and the best performing model, based on accuracy, recall, precision, and F1 Score, was used to evaluate climate change.

The second classification model used random forest due to this model's robustness against overfitting and the ability to handle correlated features. Two different models were fitted: one was fitted using the data from 1950-1960, and the other was fitted using data from 2022-2025. This set-up allowed the researchers to assess if climate patterns have shifted significantly enough that one time period cannot recognize the other. In other words, if

performance on unseen data is significantly different between both models, then that may suggest climate shifts. A model trained with data appropriate for their associated climate may not generalize well to the present-day. In contrast, a more homogenous dataset may be better at picking up subtle patterns that can more easily identify different climates.

Before training these models, monthly averages from each year were calculated and incorporated into the data frame. Features related to daily metrics were removed, resulting in a significant reduction in the datasets from 18,217 instances in the 1950-1960 dataset to 484, and 5,287 instances to 140 in the present-day dataset. Each random forest model was fitted and performance was assessed using accuracy, recall, precision, and F1 Score. Something about Euclidean Distance.

Lastly, linear regression models were created to assess if the model could predict occurrence of features such as precipitation, snowfall, and dry bulb temperature. Feature importance was assessed for each model before fitting, and the model was only fit on the datasets from 1950-1960. Performance metrics such as mean squared error, mean absolute error, and the coefficient of determination ( $R^2$ ) were used to evaluate the models.

For all models, training consisted of about 80% of the dataset and testing was the remainder. More information regarding the pipelines for these models is below in the Code section.

### III. CODE

Our project code was written in Python and organized into separate scripts for data processing, modeling, and evaluation. We used libraries such as `pandas`, `scikit-learn`, and `matplotlib`, for data handling, machine learning, and visualization respectively.

The overall pipeline includes data preprocessing, the binary classification model, the climate classification model, and the linear regression model.

#### *A. Data Preprocessing*

After extracting data from NOAA, each station had to undergo the same preprocessing steps before merging the datasets. First, the station name was replaced with the city name for clarity, and columns unrelated to this project's goal were dropped. These features included information relating to monthly averages, hourly data points, and categorical data such as report type. The features that were left related to daily values for dry bulb temperatures, precipitation, and wind.

Missing values were replaced with zeroes, as they primarily appeared to represent non-significant values. Similarly, characters such as 'T' and '\*' were inputted to signify trace amounts, such as trace snowfall or precipitation. These values were replaced with 0.005 to show that precipitation was observed, but there was not a significant amount to record. This

substitution was implemented mostly to differentiate between zeroes, indicating no precipitation, and precipitation that occurred but wasn't measurable.

Columns were converted to numeric and merged based on whether they represented a dataset from the past or present era. As stated above, the years 1950-1960 marked the past era while years 2022-2025 were included as the "present."

Across all three models, preprocessing steps were similar to one another. The 'Date' column was converted into three different features: year, month, and day. Columns that did not match between the two datasets were dropped, and the remainder were reordered so they aligned identically. Categorical data, such as city and season, were recategorized as an integer so the logistic regression and linear regression models could process the data. Lastly, a standard scaler was used on the numeric values.

For the binary classification model, an abnormal temperature column was added to be used as the target. This boolean value was calculated based on each dataset's respective quantiles for each feature and labelled "True" if any value across any feature exceeded these goalposts.

The climate classification model calculated monthly averages for each city using the daily features. These daily features were ultimately dropped, and only the newly calculated averages were used. Additionally, all rows related to 'Chicago' were dropped due to a significant number of missing values that made inclusion virtually meaningless.

### *B. Binary Classification Model*

Logistic regression was trained to identify abnormal weather based on quantile ranges from each dataset's respective era. The model was trained on the 1950-1960 dataset and tested on the present-day set. Performance metrics included precision, recall, accuracy, and F1 score. Parameter fine-tuning was conducted by assessing performance based on regularization values, starting from 0.0001 and increasing by powers of 10 to 10,000.

### *C. Climate Classification Model*

We used a random forest classifier in order to determine if climates have become more similar to one another in recent years. The model was trained on data from previous decades and tested on data from recent years. Performance was evaluated using Precision, Recall, and F1 scores as well as a confusion matrix. We then combined the train and test data for an analysis of the full timeline. By doing this we were able to calculate Euclidean distance between cities based on their climates. The idea being that if the distance between cities decreased over time, this would indicate that climates have become more similar.

### *D. Regression Model*

We implemented a series of linear regression models to predict daily precipitation, snowfall, and dry bulb temperature using historical weather features from NOAA's climate data. For each target variable, we trained a baseline model using all available features and evaluated performance using standard regression metrics: mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), and  $R^2$  score.

To improve performance, we used a Random Forest Regressor to estimate feature importance and removed weakly contributing variables. This helped reduce the dimensionality of the input space and allowed us to retrain each model with a smaller set of features.

## IV. RESULTS

For the binary classification model, accuracy, precision, recall, and F1 Score all appeared to peak at a regularization value of 100. For the training and validation set, the performance metric scores were fairly high and did not show patterns of over- or underfitting. The test set's performance was significantly worse than training. The F1 Score reached a peak of 78% in the test set compared to 86% demonstrated by the past datasets. However, performance was still above chance, indicating that the model was picking up trends that could detect abnormal weather patterns. Figure 1 shows the performance metrics across different regularization patterns in the training, validation, and test set.

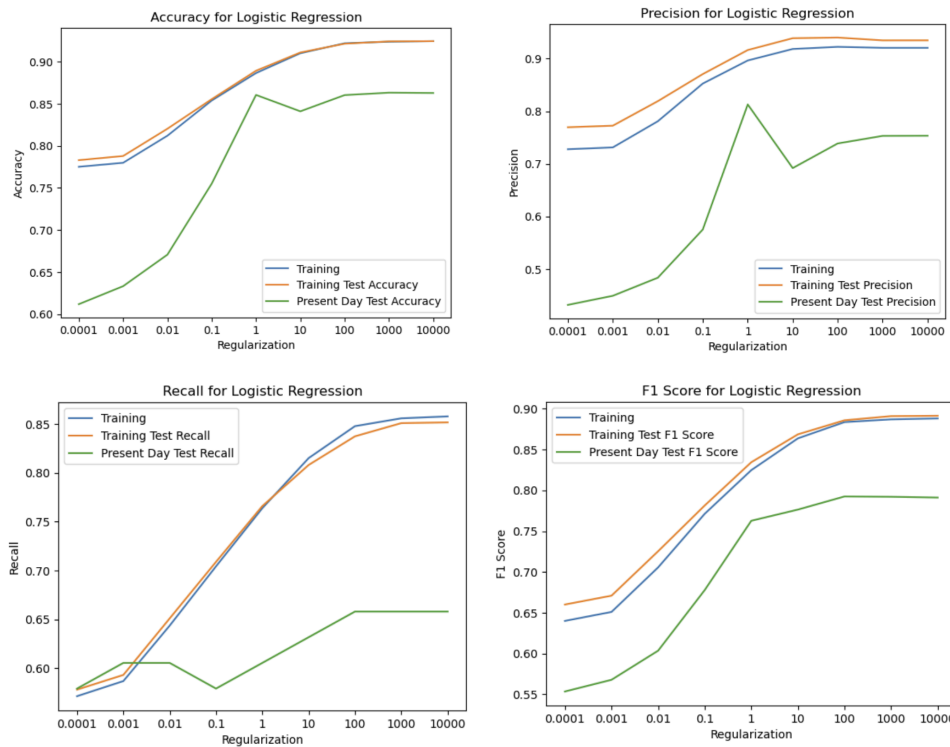


Fig. 1 Performance Metrics of Logistic Regression Model. Line graphs plot the performance of different logistic regression models with varying regularization values. The three lines represent the training set, the validation set, and the test set.

To determine if the model was picking up on climate change, a comparison between the true and predicted labels was conducted on the present-day test dataset. Overall, the model predicted a higher amount of abnormal weather days compared to the true label. Figure 2 shows the distribution of the predicted and true labels.



Fig. 2. Logistic Model Prediction of Abnormal or Normal Weather Patterns. On the x-axis, False represents normal weather patterns while True indicates abnormality. The model predicted higher incidences of abnormal weather.

The two models for the climate classification model using random forest performed very differently on the test data. ‘Chicago’ had to be removed due to the city missing key features for averaging the monthly data. For the model trained on the past data, the results appeared to be heavily biased toward classifying present-day instances as Boston. However, the model trained on the present-day data achieved better overall performance metrics when tested on the past data. Any mis-classifications seemed to be identified as Boston, but the proportion of these predictions is significantly less. The performance metrics and classification rates of these models are shown on Table 1 and Figure 3 respectively.

| Classification Report: Past Model Training Performance |           |        |          |         | Classification Report: Past Model Testing Performance |           |        |          |         |
|--------------------------------------------------------|-----------|--------|----------|---------|-------------------------------------------------------|-----------|--------|----------|---------|
|                                                        | precision | recall | f1-score | support |                                                       | precision | recall | f1-score | support |
| BOSTON                                                 | 1.00      | 1.00   | 1.00     | 121     | BOSTON                                                | 0.35      | 1.00   | 0.51     | 37      |
| MIAMI                                                  | 1.00      | 1.00   | 1.00     | 121     | MIAMI                                                 | 0.97      | 0.97   | 0.97     | 33      |
| SAN FRANCISCO                                          | 1.00      | 1.00   | 1.00     | 121     | SAN FRANCISCO                                         | 0.00      | 0.00   | 0.00     | 33      |
| WINSLOW                                                | 1.00      | 1.00   | 1.00     | 121     | WINSLOW                                               | 0.00      | 0.00   | 0.00     | 37      |
| accuracy                                               |           |        | 1.00     | 484     | accuracy                                              |           |        | 0.49     | 140     |
| macro avg                                              | 1.00      | 1.00   | 1.00     | 484     | macro avg                                             | 0.33      | 0.49   | 0.37     | 140     |
| weighted avg                                           | 1.00      | 1.00   | 1.00     | 484     | weighted avg                                          | 0.32      | 0.49   | 0.36     | 140     |

| Classification Report: Present Model Training Performance |           |        |          |         | Classification Report: Present Model Testing Performance |           |        |          |         |
|-----------------------------------------------------------|-----------|--------|----------|---------|----------------------------------------------------------|-----------|--------|----------|---------|
|                                                           | precision | recall | f1-score | support |                                                          | precision | recall | f1-score | support |
| BOSTON                                                    | 1.00      | 1.00   | 1.00     | 37      | BOSTON                                                   | 0.75      | 0.91   | 0.82     | 121     |
| MIAMI                                                     | 1.00      | 1.00   | 1.00     | 33      | MIAMI                                                    | 0.88      | 0.87   | 0.88     | 121     |
| SAN FRANCISCO                                             | 1.00      | 1.00   | 1.00     | 33      | SAN FRANCISCO                                            | 0.91      | 0.87   | 0.89     | 121     |
| WINSLOW                                                   | 1.00      | 1.00   | 1.00     | 37      | WINSLOW                                                  | 0.94      | 0.79   | 0.86     | 121     |
| accuracy                                                  |           |        | 1.00     | 140     | accuracy                                                 |           |        | 0.86     | 484     |
| macro avg                                                 | 1.00      | 1.00   | 1.00     | 140     | macro avg                                                | 0.87      | 0.86   | 0.86     | 484     |
| weighted avg                                              | 1.00      | 1.00   | 1.00     | 140     | weighted avg                                             | 0.87      | 0.86   | 0.86     | 484     |

Table 1. Performance Metrics of Climate Classification Random Forest. The bottom row represents the random forest model trained on present-day data and the top row represents the model trained on past data. The testing performance of each model is located on the right, with the set being the opposite the classifier was trained on.

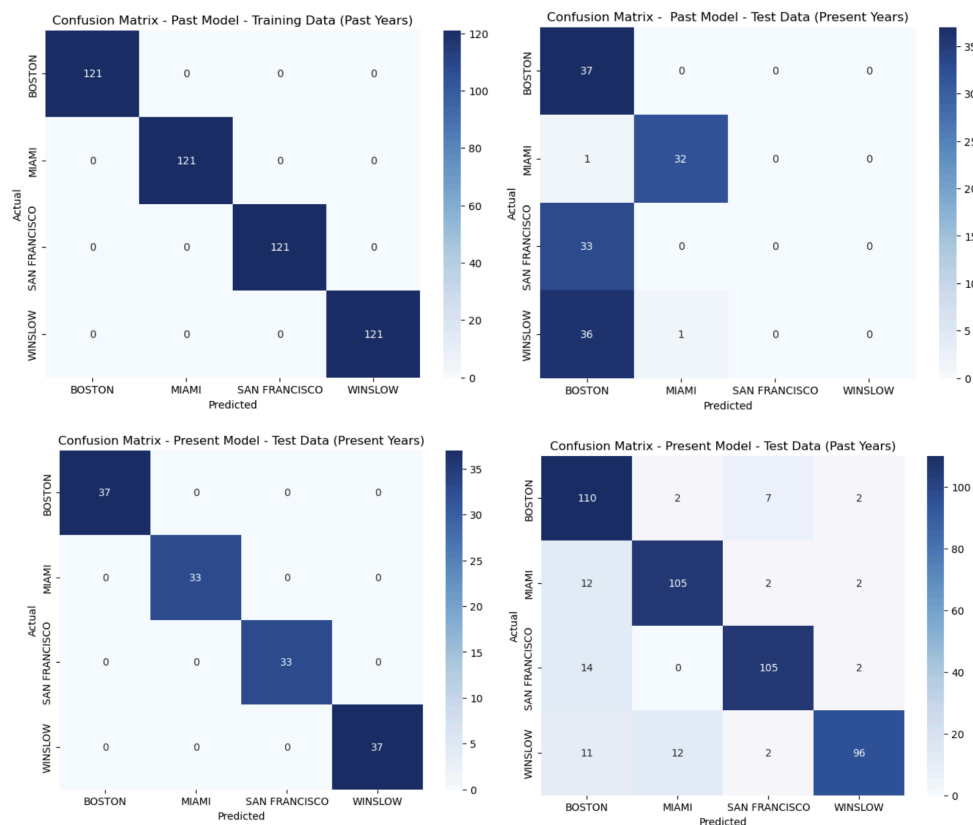


Fig 3. Correlation Matrices for Climate Classification Model. Top row shows the performance of the model that was trained on 1950 - 1960s data, and the bottom row shows present-day training. Both columns on the right show the performance of each model on an unseen test set containing data from the opposite era.

To compare the different climates, average euclidean distance was calculated, as demonstrated in Figure 4. The average distance calculated between the cities decreased in later years, suggesting the climates may be more similar to one another.



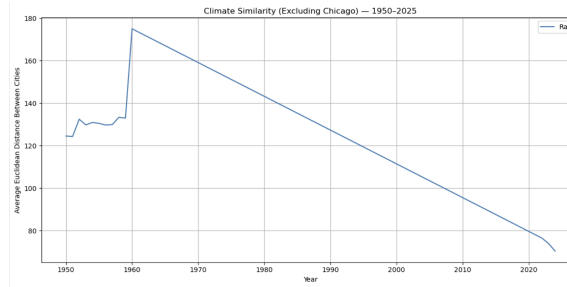


Fig 4. Euclidean Distance Over Time. Higher distance values were observed in earlier years, and smaller values in the later.

The regression model evaluated the performance of linear regression on three weather variables: precipitation, snowfall, and temperature. In the initial run using all available features, model performance was generally poor across all metrics. The precipitation model yielded an  $R^2$  of -0.086, while the snowfall model had a slightly positive  $R^2$  of 0.053. The temperature model performed the worst, with an  $R^2$  of -2.249, indicating that it explained none of the variance in the target variable.

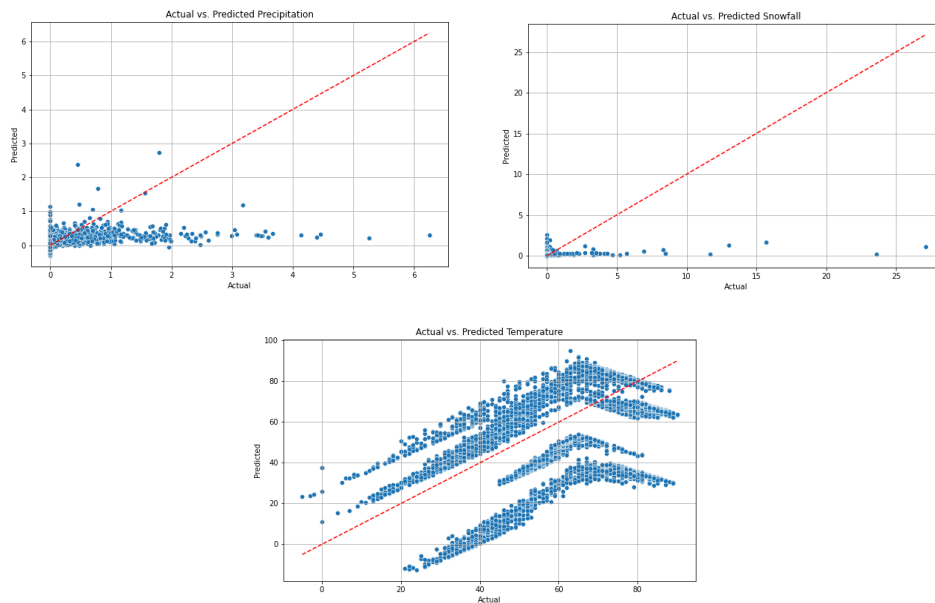


Fig 5. Linear Regression Model Performance. Chart shows the actual and predicted values for precipitation, snowfall, and dry bulb temperature.

While the random forest-based feature selection produced marginal improvements for precipitation ( $R^2 = 0.051$ ) and snowfall ( $R^2 = 0.055$ ), temperature predictions remained largely inaccurate ( $R^2 = -0.472$ ). Despite *very* slight gains, all models remained limited in their predictive power.

## V. DISCUSSION

This project created three different machine learning models in order to assess changes in climate over time in the United States. While these models performed at varying degrees of success, classification models appeared to generally outperform the linear regression model.

The logistic regression model aimed to classify instances of abnormal weather patterns based on various daily features. Training the model proved successful, performing highly across all four performance metrics scores. There appeared to be little difference between the training and validation set, which suggests that neither under- nor over-fitting occurred. Prediction on the unseen, present day dataset demonstrated comparatively worse performance, but the performance metrics stayed above chance level. As this finding indicates the model was learning the features and not choosing at random, the model appeared to be able to generalize to the present.

More frequent misclassification of normal weather as abnormal supports the theory of climate change. The label for abnormal temperature was calculated based on each past and present dataset's own quantile ranges. Because the model is trained specifically on identifying abnormal weather patterns based on the past era dataset, the classifier may be applying this standard to the present-day test set. Mis-identifying more instances as "abnormal" when they are normal for the present day could suggest that, based on the weather averages from 1950-1960, there is a higher rate of abnormal weather.

For the random forest model trained on the 1950-1960 dataset, the climate classification model had a higher rate of misclassification in recent years. Interestingly, prediction appeared to be heavily biased towards Boston. This finding could support the theory that climates are becoming more similar overtime, and this classifier is unable to generalize on more homogeneous data. If the model is trained on significantly distinct categories, generalizing on more subtle differences may prove to be more challenging. Similarly, the random forest model trained on the present-day data had a better performance rate across all performance metrics when tested against the past era dataset. These predictions could also support the idea that climates are becoming more homogeneous, as the classifier may be picking up on subtle patterns unique between different climates, allowing for a more robust model and better generalizability with more distinct categories.

To back up the idea of climate similarity, the Euclidean distance chart appears to have a negative relationship to time. A decreasing Euclidean distance indicates that climates have become increasingly similar over the last few decades. However, since the dataset excludes the years from 1960-2022, it is difficult to get a full view of what the trends were in those missing years.

The poor performance of the linear regression models gives us a limited ability to predict future temperatures and climate changes based on past trends. The consistently low  $R^2$  scores across all three linear regression models highlight the challenges of predicting daily weather outcomes using historical features alone. Even after refining the input space with Random Forest

feature selection, the models struggled to capture meaningful patterns — especially in the case of dry bulb temperature, where performance remained poor despite a drop in error metrics.

These results suggest several key takeaways. First, linear models may be too simplistic to model the nonlinear, chaotic nature of localized weather events. Daily weather is influenced by a complex interplay of atmospheric variables, many of which may not be captured in the feature set used. Second, while feature selection marginally improved performance for some targets, it was not enough to overcome the broader limitations of the modeling approach. Finally, the weak predictive power — particularly for variables like precipitation and temperature — may itself be indicative of growing unpredictability in local weather, aligning with broader concerns about climate instability and the erosion of historical norms.

## VI. FUTURE WORK

The results were limited due to the truncated dataset. For the years 1960-2022, there was no climate data available. Future work would likely include gaining access data from this time period. This way we would be able to train the models on a wider range of previous years (e.g. 1960-1990) and in turn test those models on a wider range of recent years (e.g. 1990 - present). This would better help tell the story of how climates are changing over time. Another area of focus for future projects could involve scaling the analysis globally, especially if similar datasets become available. Once this is done we could potentially correlate climate trends with policy implementations or socioeconomic impacts to better understand the impact that humans have on climate change.

## VII. CONCLUSION

This project set out to explore whether regional climate patterns in the United States have become more abnormal or homogeneous over time by applying a combination of classification and regression models to historical and recent NOAA weather data. Our findings present a nuanced view. While the classification models—particularly the climate-based random forest classifier—suggest that recent weather patterns may be converging and becoming less distinguishable from one region to another, the regression models struggled to accurately predict present-day temperature and precipitation, indicating possible nonlinear changes or previously unseen variability in the data.

These observations point toward the growing complexity of climate behavior and hint at a breakdown in the predictive power of models trained solely on historical data. This could imply that traditional expectations based on long-term averages are no longer reliable in certain regions—a finding with serious implications for planning, policy, and environmental risk assessment.

At the same time, limitations in data availability, feature consistency, and model scope present challenges to drawing definitive conclusions. For example, differences in data granularity between time periods and the relatively short window of recent data (2022–2025) may affect model robustness.

Ultimately, our analysis supports the growing consensus that climate change is not just about global warming—it also involves increasing irregularity and unpredictability at the regional level. By continuing to monitor these patterns through advanced modeling techniques, we can better prepare for the social, economic, and environmental disruptions that may lie ahead.

## VIII. REFERENCES

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## IX. ANNEX A

Much of the project was easily divisible amongst the team. Melissa handled the data preprocessing and binary classification model, Rohan took the wheel with the climate classification model, and Nikki completed the predictive linear regression model. Once the code was written, we divvied up the project report sections similarly. For the report draft, Rohan wrote the abstract, Nikki wrote the introduction, and Melissa wrote the methodology. When it came time to write the final report, we decided that the code and annex sections could be a joint effort, while Melissa extended the methodology and took on the results section, Rohan revised the abstract and wrote the discussion and future work sections, and Nikki expanded the introduction to include a literature review and wrote the conclusions and references.